Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · Agentic AI & LLM Systems · Python/Java/TypeScript · Full-Stack AI Delivery · 9yr Production

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PROFESSIONAL SUMMARY

Senior AI engineer with 9 years of production engineering depth and 2+ years of hands-on agentic AI and LLM system development. I build production-grade generative AI and agentic frameworks end-to-end: AgentProbe is an LLM evaluation framework with a 5-metric eval engine, LLM-as-judge scoring, MCP-integrated multi-provider routing (Anthropic, OpenAI, Groq, Ollama), HITL escalation, and CI-integrated regression gate -- the evaluation strategies and feedback loops this role requires. LeaseI is an 8-agent LangGraph agentic platform for document intelligence, information extraction, and cognitive automation with RAG retrieval, prompt engineering guardrails, and human-in-the-loop review -- directly demonstrating the agentic AI framework design Citi's banking AI squads need. Claude Code is my primary development environment (used daily with MCP integration). At Entrata, I owned compliance-critical production systems processing millions of regulated records annually -- 7 years of building software where wrong data has real consequences, the same discipline regulated financial AI demands. Python (9yr), Java, TypeScript, AWS, Kubernetes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI & Agentic Systems: LangGraph, LangChain, Agentic AI, Multi-Agent Systems, LLM Orchestration, Tool Calling, MCP (Model Context Protocol), RAG Pipelines, Embeddings, Vector Databases (ChromaDB), Semantic Search, Prompt Engineering, Few-Shot & Chain-of-Thought Prompting, Output Validation, Guardrails, HITL (Human-in-the-Loop), LLM Evaluation, LLM-as-Judge, Hallucination Detection, Agent Observability, Knowledge Representation; LLM APIs: Anthropic/Claude, OpenAI, Groq, Ollama; Claude Code (daily), GitHub Copilot

Languages & Frameworks: Python (9yr production), Java/Spring Boot, TypeScript, JavaScript, Angular, Node.js, SQL/PL-SQL, PHP, R, Bash, HTML/CSS; FastAPI, Flask, REST APIs, React.js; TensorFlow, PyTorch, scikit-learn, XGBoost, NumPy, Pandas, spaCy, BERT, Hugging Face; data engineering, ETL/ELT pipelines, microservices, event-driven architecture, API-first design

Cloud, Infrastructure & Practices: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, OpenShift (K8s-compatible), CI/CD (GitHub Actions), Git, PostgreSQL, MySQL, Oracle, NoSQL; SDLC, Agile/Scrum, Code Reviews, Security & Compliance, Performance Optimization, Observability; scalable distributed systems, fault-tolerant backend services

EXPERIENCE

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Systems & Generative AI

- Built an agentic AI mission-planning system as a LangGraph stateful graph with multi-step planning, RAG retrieval over a regulatory corpus (FAA Part 107) with cited sources, and validation guardrails enforcing compliance constraints before export -- demonstrating the MVP-first, rapid-iteration agentic delivery pattern Citi's banking AI squads require.
- Engineered prompt engineering standards and output validation logic; designed conditional routing with graceful failure handling for non-compliant outputs with grounded rejection reasoning.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest anomaly detection across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and data engineering.

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- AI/Data Engineering, Compliance Systems & Backend Services · Lehi, UT (Remote)

- **Compliance-critical production systems in Python, Java, and PostgreSQL.** Owned Entrata's annual HUD income-limit pipeline for 7 years: federal regulatory data transformed and mapped to ~180,000 domain records powering eligibility and rent calculations for 100K+ housing units. Zero compliance violations across 7 years. This is the same engineering discipline that regulated financial AI demands -- systems where wrong data has real consequences.
- **Scalable, reliable backend services and distributed architecture.** RabbitMQ-based async processing with idempotency guarantees; optimized 200+ SQL queries (30% performance gain) through execution plan analysis and stored procedure refactoring; Docker microservices redesign (50% scalability improvement); AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance); 99.5% uptime across 20+ full-stack production modules.
- **Full-stack engineering across Python, Java, PHP, JavaScript, Angular.** Shipped 20+ full-stack production modules across 120+ Agile sprints; built internal AI-assisted debugging tools reducing debugging time 40%; integrated third-party APIs and vendor data platforms; defined CI/CD and code quality standards across the team.
- **Technical leadership and mentorship.** Mentored 5+ engineers; conducted 300+ peer code reviews; championed iterative delivery, observability, and security practices; maintained bug count below 10 for 6+ consecutive months.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · Java · Spring Boot · PostgreSQL

- Built compliance modules in Python, Java, and Spring Boot; income-limit import pipeline (1.6M HUD/Tax Credit records); AWS S3 migration; database schema redesign with post-deployment stabilization.

PROJECTS

AgentProbe: Production LLM Evaluation & Agentic AI Framework

LLM Evaluation · MCP · Multi-Provider · HITL · CI/CD · Agentic AI

Python · LangGraph · FastAPI · ChromaDB · React.js · TypeScript · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation framework demonstrating the evaluation strategies and feedback loops Citi's AI squads need: 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-judge scoring with multi-trial pass@k harness, MCP-integrated multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), CI-integrated regression gate blocking quality degradation, HITL escalation, structured agent observability and tracing. Used to evaluate LeaseI across 1,240+ test cases. Claude Code is the development environment -- same tool Citi named in this JD.

LeaseI: Agentic AI Platform for Cognitive Automation

LangGraph · Agentic AI · GenAI · RAG · Prompt Engineering · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end generative AI and agentic framework for cognitive automation: information extraction, summarization, and question answering from large-scale regulated documents. 8-agent LangGraph pipeline with conditional routing, OCR-fallback ingestion, TF-IDF RAG retrieval, prompt engineering guardrails, and HITL escalation on regulated decisions. 93.6% precision, 87.9% F1 across 1,240 test cases (5 US states + federal regulatory corpus). Demonstrates the agentic AI systems and iterative delivery approach Citi's Banking Technology squads require.

AI on FHIR: Secure Health Data Platform -- HIPAA & Regulated Compliance

Flask · REST API · HIPAA · RBAC · NLP · Financial-Grade Security

Python · Flask · React · Docker · FHIR API · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant AI platform integrating NLP (BERT, spaCy) with federal health data APIs; RBAC security controls and governance for AI-generated outputs. Demonstrates the security, compliance, and regulated environment engineering that banking AI solutions require.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017